

Independence of two random variables

Definition (Independence (two random variables))

Let X and Y be two random variables defined in a probability space with ranges T_X and T_Y , respectively. X and Y are said to be independent if any event defined using X alone is independent of any event defined using Y alone. Equivalently, if the joint PMF of X and Y is f_{XY} , X and Y are independent if

$$f_{XY}(t_1, t_2) = f_X(t_1)f_Y(t_2)$$

for $t_1 \in T_X$, $t_2 \in T_Y$.

- General case vs independent case
 - ▶ General: $f_{XY}(t_1, t_2) = f_X(t_1)f_{Y|X=t_1}(t_2)$ (always true)
 - ▶ Independent: $f_{Y|X=t_1}(t_2) = f_Y(t_2)$
- If X and Y are independent
 - ▶ Joint PMF equals product of marginal PMFs
 - ▶ Conditional PMF equals marginal PMF

Simple examples

Independent

$t_2 \setminus t_1$	0	1	f_Y
0	1/4	1/4	1/2
1	1/4	1/4	1/2
f_X	1/2	1/2	

$$f_{X,Y}(t_1, t_2) = f_X(t_1)f_Y(t_2) \text{ for all } t_1, t_2$$

Simple examples

$t_2 \setminus t_1$	0	1	f_Y
0	$1/4$	$1/4$	$1/2$
1	$1/4$	$1/4$	$1/2$
f_X	$1/2$	$1/2$	

indep

NOT INDEPENDENT

$t_2 \setminus t_1$	0	1	f_Y
0	$1/2$	$1/4$	$3/4$
1	$1/8$	$1/8$	$1/4$
f_X	$5/8$	$3/8$	

$$f_{XY}(0,0) = 1/2 \neq \underset{5/8}{f_X(0)} \cdot \underset{3/4}{f_Y(0)}$$

Simple examples

$t_2 \setminus t_1$	0	1	f_Y
0	1/4	1/4	
1	1/4	1/4	
f_X	<u>indep</u>		

$t_2 \setminus t_1$	0	1	f_Y
0	1/2	1/4	
1	1/8	1/8	
f_X	<u>not indep</u>		

$t_2 \setminus t_1$	0	1	f_Y
0	x	$1/2 - x$	$1/2$
1	$1/2 - x$	x	$1/2$
f_X	$1/2$	$1/2$	

if $x = 1/4$, indep
 $x \neq 1/4$, not indep.

More examples

$t_2 \setminus t_1$	0	1	2	f_Y
0	$1/9$	$1/9$	$1/9$	$1/3$
1	$1/9$	$1/9$	$1/9$	$1/3$
2	$1/9$	$1/9$	$1/9$	$1/3$
f_X	$1/3$	$1/3$	$1/3$	

indep

More examples

$t_2 \setminus t_1$	0	1	2	f_Y
0	$1/9$	$1/9$	$1/9$	
1	$1/9$	$1/9$	$1/9$	
2	$1/9$	$1/9$	$1/9$	
f_X		<u>indep</u>		

$t_2 \setminus t_1$	0	1	2	f_Y
0	$1/6$	$1/12$	$1/12$	$1/3$
1	$1/4$	$1/8$	$1/8$	$1/2$
2	$1/12$	$1/24$	$1/24$	$1/6$
f_X	$1/2$	$1/4$	$1/4$	

indep

More examples

$t_2 \setminus t_1$	0	1	2	f_Y
0	1/9	1/9	1/9	
1	1/9	1/9	1/9	
2	1/9	1/9	1/9	
f_X				

indep

$t_2 \setminus t_1$	0	1	2	f_Y
0	1/6	1/12	1/12	
1	1/4	1/8	1/8	
2	1/12	1/24	1/24	
f_X				

indep

$t_2 \setminus t_1$	0	1	2	f_Y
0	0	1/8	1/8	1/4
1	1/8	1/8	1/8	3/8
2	1/8	1/8	1/8	3/8
f_X	1/4	3/8	3/8	

dependent

More examples

$t_2 \setminus t_1$	0	1	2	f_Y
0	1/9	1/9	1/9	
1	1/9	1/9	1/9	
2	1/9	1/9	1/9	
f_X		indep		

$t_2 \setminus t_1$	0	1	2	f_Y
0	1/6	1/12	1/12	
1	1/4	1/8	1/8	
2	1/12	1/24	1/24	
f_X		indep		

$t_2 \setminus t_1$	0	1	2	f_Y
0	0	1/8	1/8	
1	1/8	1/8	1/8	
2	1/8	1/8	1/8	
f_X		dep		

$t_2 \setminus t_1$	0	1	2	f_Y
0	1/6	1/12	1/8	
1	1/4	1/8	1/8	
2	1/24	1/24	1/24	
f_X				

Handwritten annotations: A circle around the value 1/8 in the cell (1, 2). An arrow points from this circle to the value 1/24 in the cell (2, 2). Another arrow points from the circled 1/8 to the value 1/24 in the cell (2, 1). The word "dep" is written below the table.

Independent vs dependent random variables

- To show X and Y are independent, verify

$$f_{XY}(t_1, t_2) = f_X(t_1)f_Y(t_2)$$

for **all** $t_1 \in T_X$, $t_2 \in T_Y$.

- To show X and Y are dependent, verify

$$f_{XY}(t_1, t_2) \neq f_X(t_1)f_Y(t_2)$$

for **some** $t_1 \in T_X$ and $t_2 \in T_Y$.

- ▶ Special case: $f_{XY}(t_1, t_2) = 0$ when $f_X(t_1) \neq 0$, $f_Y(t_2) \neq 0$

Example: Random 2-digit number

A 2-digit number from 00 to 99 is selected at random. Partial information is available about the number as two random variables. Let X be the digit in units place. Let Y be the remainder obtained when the number is divided by 4.

$$X \sim \text{Uniform}\{0, 1, 2, \dots, 9\}$$

$$Y \sim \text{Uniform}\{0, 1, 2, 3\}$$

$$f_{XY}(1, 0) = 0 \neq f_X(1) \cdot f_Y(0)$$

$$\Rightarrow X \neq Y: \text{dependent}$$

Example: IPL powerplay over

Let X = number of runs in the over. Let Y = number of wickets in the over.

Are X & Y independent?

Independence of multiple random variables

Definition (Independence of random variables)

Let $X_1, \dots, X_n$ be random variables defined in a probability space with range of X_i denoted T_{X_i} . $X_1, \dots, X_n$ are said to be independent if events defined using different X_i are mutually independent. Equivalently, $X_1, \dots, X_n$ are independent iff

$$f_{X_1 \dots X_n}(t_1, \dots, t_n) = f_{X_1}(t_1)f_{X_2}(t_2) \cdots f_{X_n}(t_n)$$

for all $t_i \in T_{X_i}$.

- If X and Y are independent
 - ▶ Joint PMF equals product of marginal PMFs
 - ▶ Conditional PMF equals unconditioned PMF
- All subsets of independent random variables are independent

Example: Toss a fair coin thrice

Let $X_i = 1$ if i -th toss is heads and $X_i = 0$ if i -th toss is tails, $i = 1, 2, 3$.

Joint PMF: $f_{X_1 X_2 X_3}(t_1, t_2, t_3) = 1/8$ for $t_i \in \{0, 1\}$

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Joint PMF: $f_{X_1 X_2 X_3}(t_1, t_2, t_3) = 1/8$ for $t_i \in \{0, 1\}$

- $X_i \sim \text{Uniform}\{0, 1\}$

$$f_{X_1 X_2 X_3}(t_1, t_2, t_3) = f_{X_1}(t_1) f_{X_2}(t_2) f_{X_3}(t_3)$$

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- $X_i \sim \text{Uniform}\{0, 1\}$

$$f_{X_1 X_2 X_3}(t_1, t_2, t_3) = f_{X_1}(t_1) f_{X_2}(t_2) f_{X_3}(t_3)$$

X_1, X_2, X_3 : Independent random variables

Example: Random 3-digit number 000 to 999

X = first digit from left, Y = number modulo 2, Z = first digit from right

- $X \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $Y \sim \text{Uniform}\{0, 1\}$
- $Z \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

Example: Random 3-digit number 000 to 999

X = first digit from left, Y = number modulo 2, Z = first digit from right

- $X \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $Y \sim \text{Uniform}\{0, 1\}$
- $Z \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- X, Z
 - ▶ $f_{XZ}(t_1, t_3) = 1/100 = f_X(t_1)f_Z(t_3)$
 - ▶ Independent
- X, Y
 - ▶ $f_{XY}(t_1, t_2) = 1/20 = f_X(t_1)f_Y(t_2)$
 - ▶ Independent
- Y, Z
 - ▶ $f_{YZ}(1, 2) = 0 \neq f_Y(1)f_Z(2)$
 - ▶ Not independent

Example: Random 3-digit number 000 to 999

X = first digit from left, Y = number modulo 2, Z = first digit from right

- $X \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- $Y \sim \text{Uniform}\{0, 1\}$
- $Z \sim \text{Uniform}\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- X, Z
 - ▶ $f_{XZ}(t_1, t_3) = 1/100 = f_X(t_1)f_Z(t_3)$
 - ▶ Independent
- X, Y
 - ▶ $f_{XY}(t_1, t_2) = 1/20 = f_X(t_1)f_Y(t_2)$
 - ▶ Independent
- Y, Z
 - ▶ $f_{YZ}(1, 2) = 0 \neq f_Y(1)f_Z(2)$
 - ▶ Not independent

X, Y, Z : Dependent random variables

Example: Even parity

t_1	t_2	t_3	$f_{X_1 X_2 X_3}(t_1, t_2, t_3)$
0	0	0	1/4
0	1	1	1/4
1	0	1	1/4
1	1	0	1/4

- Number of 1s in (X_1, X_2, X_3) is even (hence, the name)

Example: Even parity

t_1	t_2	t_3	$f_{X_1 X_2 X_3}(t_1, t_2, t_3)$
0	0	0	1/4
0	1	1	1/4
1	0	1	1/4
1	1	0	1/4

- Number of 1s in (X_1, X_2, X_3) is even (hence, the name)
- $X_i \sim \text{Uniform}\{0, 1\}$
- All pairs are independent
- $f_{X_1 X_2 X_3}(0, 0, 1) = 0 \neq f_{X_1}(0)f_{X_2}(0)f_{X_3}(1)$

Example: Even parity

t_1	t_2	t_3	$f_{X_1 X_2 X_3}(t_1, t_2, t_3)$
0	0	0	1/4
0	1	1	1/4
1	0	1	1/4
1	1	1	1/4

- Number of 1s in (X_1, X_2, X_3) is even (hence, the name)
- $X_i \sim \text{Uniform}\{0, 1\}$
- All pairs are independent
- $f_{X_1 X_2 X_3}(0, 0, 1) = 0 \neq f_{X_1}(0)f_{X_2}(0)f_{X_3}(1)$

X_1, X_2, X_3 : Dependent random variables

Independent and Identically Distributed (i.i.d.)

Definition (i.i.d.)

Random variables $X_1, \dots, X_n$ are said to be independent and identically distributed (i.i.d.), if

- 1 they are independent,
- 2 the marginal PMFs f_{X_i} are identical.

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- ① they are independent,
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 - ▶ Toss a coin multiple times
 - ▶ Throw a die multiple times

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- Repeated trials of an experiment creates i.i.d. sequence of random variables
 - ▶ Toss a coin multiple times
 - ▶ Throw a die multiple times

Notation: $X_1, X_2, \dots, X_n$ are i.i.d. with distribution X means that f_{X_i} is the same as f_X

Handwritten notation showing a box containing $X_1, X_2, \dots, X_n \sim \text{iid } X$. An arrow points from the X to the PMF f_X . Below the box, it says $X_1, X_2, \dots, X_n \sim \text{iid } f_X$ with a small 'con' above the tilde.

Problem: i.i.d. Geometric

Let $X_1, \dots, X_n$ be i.i.d. with a Geometric(p) distribution. What is the probability that all of these random variables are larger than some positive integer j ?

$$X \sim \text{Geometric}(p) : X \in \left\{ 1, 2, 3, \dots \right\} \quad P(X=k) = (1-p)^{k-1} p$$

$$P(X_1 > j, X_2 > j, \dots, X_n > j) = P(X_1 > j) P(X_2 > j) \dots P(X_n > j) \\ = \left(P(X > j) \right)^n = (1-p)^{jn}$$

$$P(X > j) = \sum_{k=j+1}^{\infty} (1-p)^{k-1} p = \frac{(1-p)^j p}{1-(1-p)} = (1-p)^j$$

Problem: i.i.d. samples

Let $X \sim \left\{ 0, 1, 2, 3, 4 \right\}$ with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}$ respectively, and let $X_1, \dots, X_n$ be i.i.d. samples with distribution X .

- What is the probability that 4 is missing in the samples?
- What is the probability that 4 appears exactly once in the samples?
- What is the probability that 3 and 4 appear at least once in the samples?

$$\textcircled{1} P(X_1 \neq 4, X_2 \neq 4, \dots, X_n \neq 4) = (P(X \neq 4))^n = \left(\frac{15}{16}\right)^n$$

$$\textcircled{2} P(4 \text{ appears exactly once}) = P(X_1 = 4, X_2 \neq 4, \dots, X_n \neq 4) + P(X_1 \neq 4, X_2 = 4, X_3 \neq 4, \dots, X_n \neq 4) + \dots + P(X_1 \neq 4, \dots, X_{n-1} \neq 4, X_n = 4) = n P(X=4) (P(X \neq 4))^{n-1} = n \left(\frac{1}{16}\right) \left(\frac{15}{16}\right)^{n-1}$$

$$\textcircled{3} P(3 \text{ at least once} \cap 4 \text{ at least once})$$

$$A \cap B = (A^c \cup B^c)^c$$

$$P(A^c) = \left(\frac{15}{16}\right)^n, P(B^c) = \left(\frac{15}{16}\right)^n, P(A^c \cap B^c) = \left(\frac{14}{16}\right)^n \Rightarrow P(A^c \cup B^c) = 2\left(\frac{15}{16}\right)^n - \left(\frac{14}{16}\right)^n$$

$$P(A \cap B) = 1 - \left(2\left(\frac{15}{16}\right)^n - \left(\frac{14}{16}\right)^n\right)$$

Problem: Memoryless property of Geometric

Let $X \sim \text{Geometric}(p)$. Find the following.

① $P(X > n)$

② $P(X > m + n | X > m)$

$$\begin{aligned} \textcircled{1} \quad P(X > n) &= \sum_{k=n+1}^{\infty} (1-p)^{k-1} p = (1-p)^n p + (1-p)^{n+1} p + \dots \\ &= \frac{(1-p)^n p}{1 - (1-p)} = (1-p)^n \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad P(X > m+n | X > m) &= \frac{P(X > m+n \cap X > m)}{P(X > m)} = \frac{P(X > m+n)}{P(X > m)} \\ &= \frac{(1-p)^{m+n}}{(1-p)^m} = (1-p)^n \end{aligned}$$

$$P(X > m+n | X > m) = P(X > n)$$

Visualizing functions of random variables

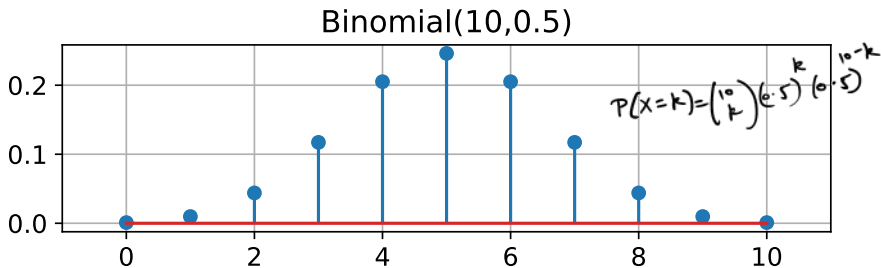
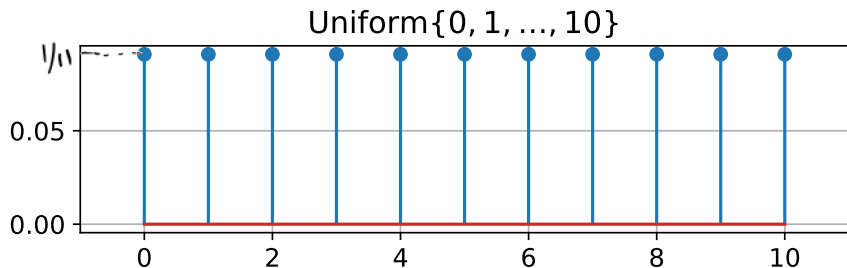
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Section 1

One random variable, one-to-one functions

Uniform $\{0, 1, \dots, 10\}$ and Binomial(10,0.5): PMF plots



PMF of $Y = X - 5$ (table method) $Y \in \{-5, -4, \dots, 5\}$

Uniform $\{0, 1, \dots, 10\}$

one-to-one

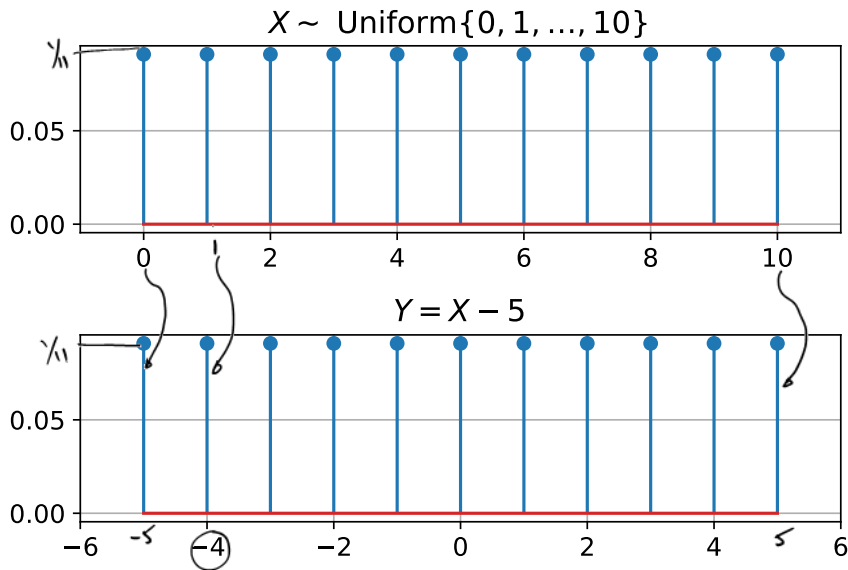
x	$P(X = x)$	$y = x - 5$
0	1/11	-5
1	1/11	-4
2	1/11	-3
3	1/11	-2
4	1/11	-1
5	1/11	0
6	1/11	1
7	1/11	2
8	1/11	3
9	1/11	4
10	1/11	5

no repetition

Binomial(10,0.5)

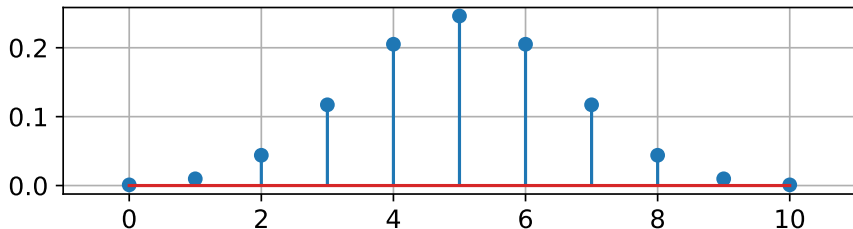
x	$P(X = x)$	$y = x - 5$
0	0.00097656	-5
1	0.00976563	-4
2	0.04394531	-3
3	0.1171875	-2
4	0.20507813	-1
5	0.24609375	0
6	0.20507813	1
7	0.1171875	2
8	0.04394531	3
9	0.00976563	4
10	0.00097656	5

PMF plots of $X \sim \text{Uniform}\{0, 1, \dots, 10\}$ and $Y = X - 5$

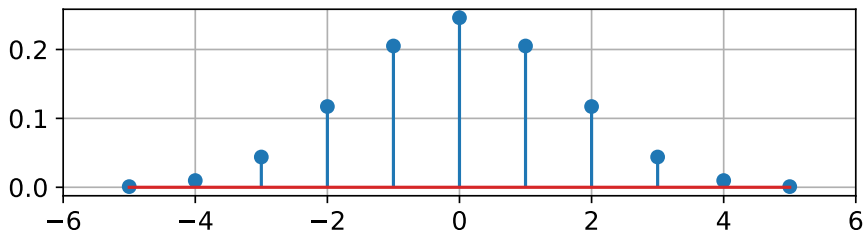


PMF plots of $X \sim \text{Binomial}(10, 0.5)$ and $Y = X - 5$

$X \sim \text{Binomial}(10, 0.5)$



$Y = X - 5$



PMF of $Y = 2^X$ (table method)

Uniform $\{0, 1, \dots, 10\}$

one-to-one

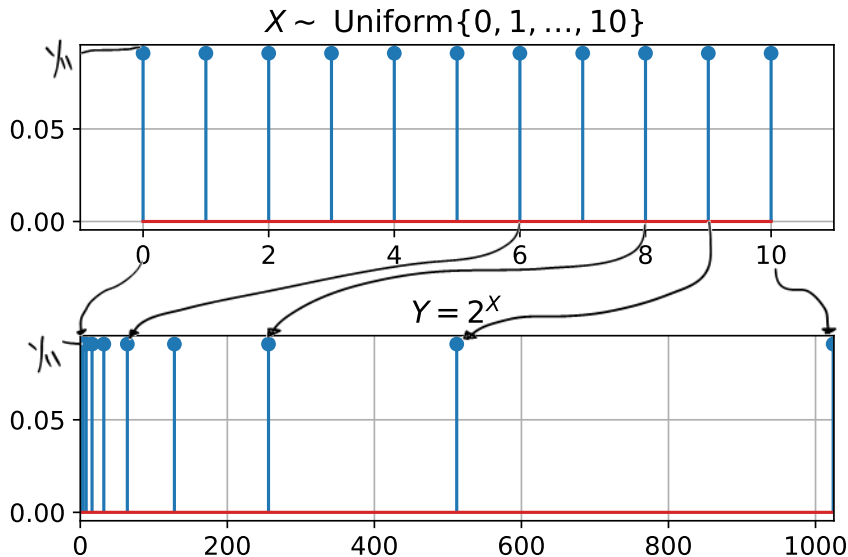
x	$P(X = x)$	$y = 2^x$
0	1/11	1
1	1/11	2
2	1/11	4
3	1/11	8
4	1/11	16
5	1/11	32
6	1/11	64
7	1/11	128
8	1/11	256
9	1/11	512
10	1/11	1024

no repetition

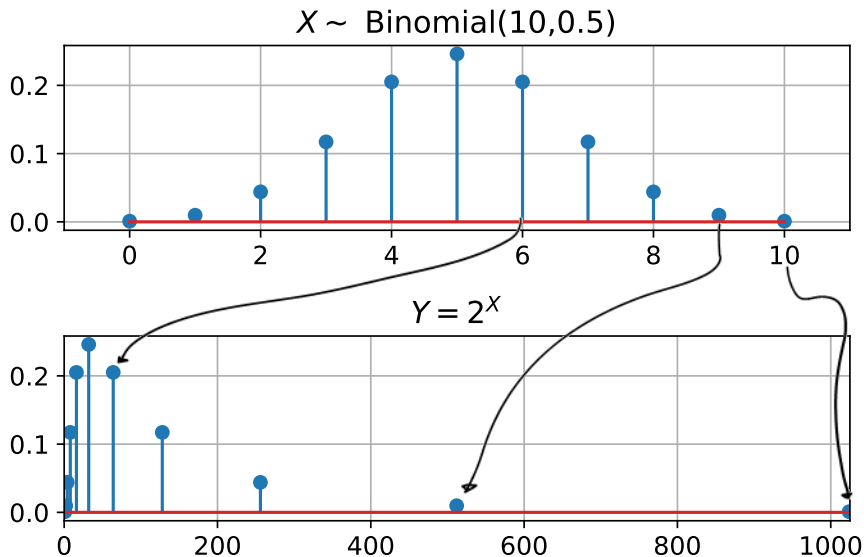
Binomial(10,0.5)

x	$P(X = x)$	$y = 2^x$
0	0.00097656	1
1	0.00976563	2
2	0.04394531	4
3	0.1171875	8
4	0.20507813	16
5	0.24609375	32
6	0.20507813	64
7	0.1171875	128
8	0.04394531	256
9	0.00976563	512
10	0.00097656	1024

PMF plots of $X \sim \text{Uniform}\{0, 1, \dots, 10\}$ and $Y = 2^X$



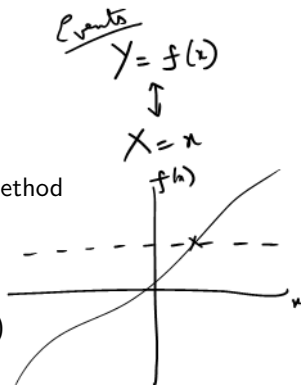
PMF plots of $X \sim \text{Binomial}(10, 0.5)$ and $Y = 2^X$



One-to-one functions

- One-to-one: For different input, the function results in different outputs
 - ▶ In the plot of $y = f(x)$ versus x , every horizontal line intersects the curve only at one point.
 - ▶ Monotonic functions are one-to-one
 - ▶ Example: $x - 5$, 2^x
- If f is one-to-one
 - ▶ Easy to find PMF of $f(X)$ using the table method
 - ▶ Easy to visualize the plot of the PMF

$$P(Y = f(x)) = P(X = x)$$



Section 2

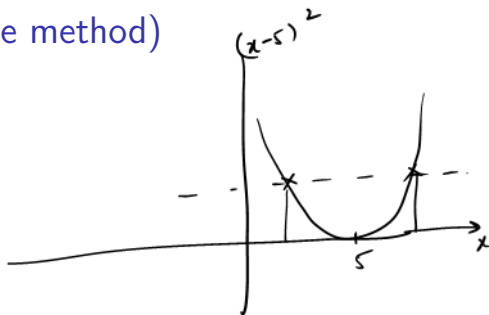
One random variable, many-to-one functions

PMF of $Y = (X - 5)^2$ (table method)

$X \sim \text{Uniform}\{0, 1, \dots, 10\}$

x	$P(X = x)$	$y = (x - 5)^2$
0	1/11	25
1	1/11	16
2	1/11	9
3	1/11	4
4	1/11	1
5	1/11	0
6	1/11	1
7	1/11	4
8	1/11	9
9	1/11	16
10	1/11	25

→ have repetitions



PMF of $Y = (X - 5)^2$ (table method)

$X \sim \text{Uniform}\{0, 1, \dots, 10\}$

$$Y = (X - 5)^2$$

x	$P(X = x)$	$y = (x - 5)^2$	y	$P(Y = y)$
0	1/11	25	0	1/11
1	1/11	16	1	2/11
2	1/11	9	4	2/11
3	1/11	4	9	2/11
4	1/11	1	16	2/11
5	1/11	0	25	2/11
6	1/11	1		
7	1/11	4		
8	1/11	9		
9	1/11	16		
10	1/11	25		

$Y \in \{0, 1, 4, 9, 16, 25\}$

PMF of $Y = (X - 5)^2$ (table method)

$X \sim \text{Uniform}\{0, 1, \dots, 10\}$

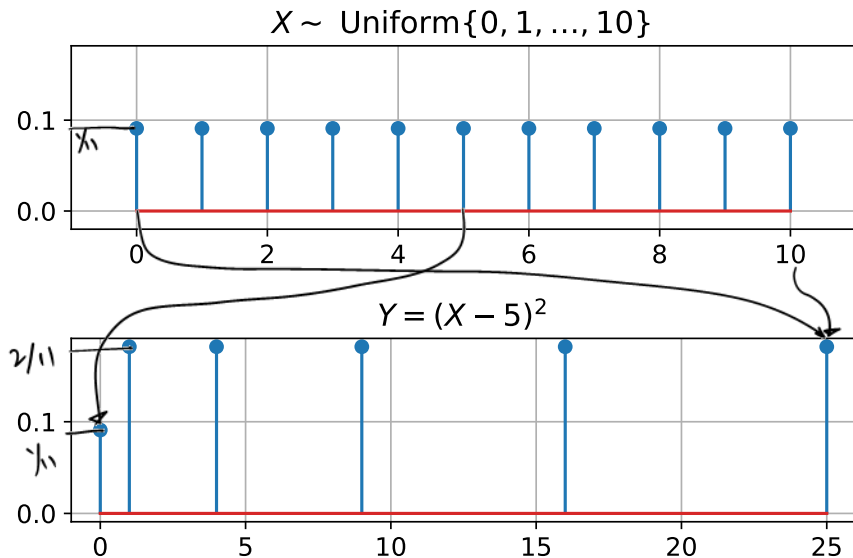
$Y = (X - 5)^2$

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1	1/11	16
2	1/11	9
3	1/11	4
4	1/11	1
5	1/11	0
6	1/11	1
7	1/11	4
8	1/11	9
9	1/11	16
10	1/11	25

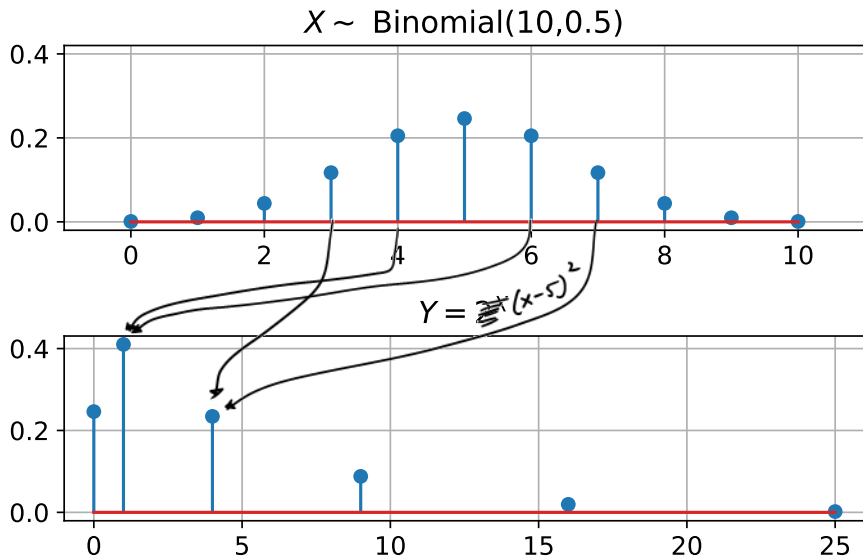
y	$P(Y = y)$
0	1/11
1	2/11
4	2/11
9	2/11
16	2/11
25	2/11

- $(x - 5)^2$ is **not** one-to-one

PMF plots: $X \sim \text{Uniform}\{0, 1, \dots, 10\}$, $Y = (X - 5)^2$



PMF plots: $X \sim \text{Binomial}(10, 0.5)$, $Y = (X - 5)^2$



Many-to-one functions

- Many-to-one: For two or more different input, the function should result in the same output
 - ▶ In the plot of $y = f(x)$ versus x , some horizontal line intersects the curve at more than one point.
 - ▶ *Non-monotonic* functions are many-to-one
 - ▶ Example: $y = x^2$, $y = x(1 - x)$, $y = xe^{-x}$
- If f is many-to-one
 - ▶ The table method will generally work. . . .
 - ▶ Let y_0 be a value taken by f at the points $x_1, x_2, \dots, x_m$ and nowhere else. That is, $y_0 = f(x_1) = \dots = f(x_m)$.

$$\underline{\text{Events:}} \{Y = y_0\} = \{X \in \{x_1, \dots, x_m\}\}$$

$$P(Y = y_0) = P(X = x_1) + \dots + P(X = x_m)$$

Problem

Let $X \sim \text{Uniform}\{-5, -4, \dots, 5\}$. Let

$$f(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0. \end{cases}$$

Find the distribution of $Y = f(X)$.

x	$P(X=x)$	$y=f(x)$
-5	$1/11$	0
-4	$1/11$	0
-3	$1/11$	0
-2	$1/11$	0
-1	$1/11$	0
0	$1/11$	0
1	$1/11$	1
2	$1/11$	2
3	$1/11$	3
4	$1/11$	4
5	$1/11$	5

$y \in \{0, 1, 2, 3, 4, 5\}$	
y	$P(Y=y)$
0	$6/11$
1	$1/11$
2	$1/11$
3	$1/11$
4	$1/11$
5	$1/11$

Problem

Let $X \sim \text{Uniform}\{-500, -499, \dots, 500\}$. Let $f(x) = \max(x, 5)$. Find the distribution of $Y = f(X)$.

$$= \begin{cases} x, & x \geq 5 \\ 5, & x < 5 \end{cases}$$

x	$P(X=x)$	y
-500	$1/1001$	5
$\vdots$	$\vdots$	$\vdots$
4	$1/1001$	5
5	$1/1001$	5
6	$1/1001$	6
$\vdots$	$\vdots$	$\vdots$
500	$1/1001$	500

506 entries

$$Y \in \{5, 6, 7, \dots, 500\}$$

y	$P(Y=y)$
5	$506/1001$
6	$1/1001$
7	$1/1001$
$\vdots$	$\vdots$
500	$1/1001$

495 entries

Section 3

Two random variables

Small examples and table method: Sum

$$X, Y \sim \text{iid Unif}\{0, 1\}, Z = X + Y$$

x	y	$f_{XY}(x, y)$	z
0	0	1/4	0
0	1	1/4	1
1	0	1/4	1
1	1	1/4	2

↓
rejection

Small examples and table method: Sum

$$X, Y \sim \text{iid Unif}\{0, 1\}, Z = X + Y$$

x	y	$f_{XY}(x, y)$	z
0	0	1/4	0
0	1	1/4	1
1	0	1/4	1
1	1	1/4	2

$$Z \in \{0, 1, 2\}$$

Small examples and table method: Sum

$$X, Y \sim \text{iid Unif}\{0, 1\}, Z = X + Y$$

x	y	$f_{XY}(x, y)$	z
0	0	1/4	0
0	1	1/4	1
1	0	1/4	1
1	1	1/4	2

$$Z \in \{0, 1, 2\}$$

$$P(Z = 0) = 1/4, P(Z = 1) = 1/2, P(Z = 2) = 1/4$$

Small examples and table method: Max

$Z = \max(X, Y)$

joint PDF of $X+Y$

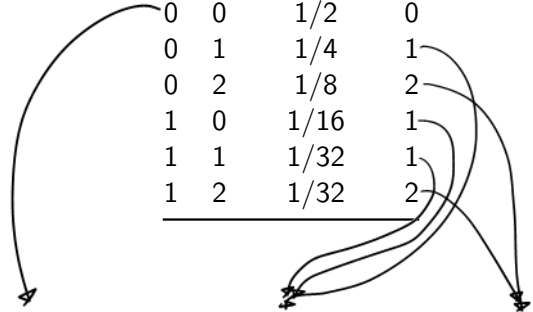
x	y	$f_{XY}(x, y)$	Z
0	0	$1/2$	0
0	1	$1/4$	1
0	2	$1/8$	2
1	0	$1/16$	1
1	1	$1/32$	1
1	2	$1/32$	2

$x \backslash y$	0	1	2
0	$1/2$	$1/4$	$1/8$
1	$1/16$	$1/32$	$1/32$

Small examples and table method: Max

$$Z = \max(X, Y)$$

x	y	$f_{XY}(x, y)$	z
0	0	$1/2$	0
0	1	$1/4$	1
0	2	$1/8$	2
1	0	$1/16$	1
1	1	$1/32$	1
1	2	$1/32$	2


$$P(Z = 0) = 1/2, P(Z = 1) = 11/32, P(Z = 2) = 5/32$$

$$\frac{1}{4} + \frac{1}{16} + \frac{1}{32} =$$

$$\frac{1}{8} + \frac{1}{32}$$

Moderate size examples: Too cumbersome

- Pair of fair dice are thrown. What is the distribution of the sum or max or min?

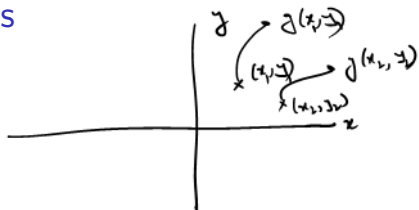
Moderate size examples: Too cumbersome

- Pair of fair dice are thrown. What is the distribution of the sum or max or min?
- Table method
 - ▶ 36 possible values
 - ▶ Prone to errors

Moderate size examples: Too cumbersome

- Pair of fair dice are thrown. What is the distribution of the sum or max or min?
- Table method
 - ▶ 36 possible values
 - ▶ Prone to errors
- What if size becomes 100?

Visualizing functions of 2 variables



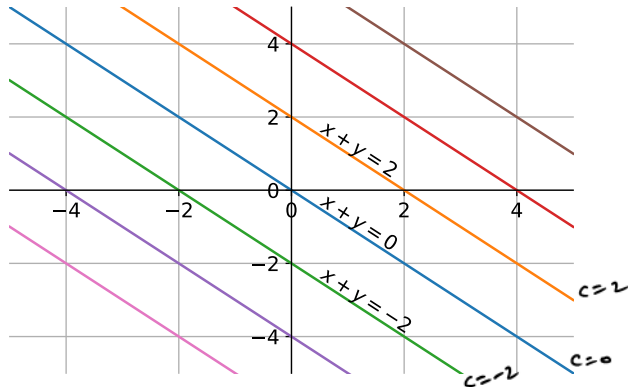
- $g(x, y)$: function
 - ▶ 3D plot is one option, but often not very useful
- Contours I: values of (x, y) that result in $g(x, y) = c$
 - ▶ Make a plot of those (x, y) for different c
- ^{Regions}~~Contours~~ II: values of (x, y) that result in $g(x, y) \leq c$
 - ▶ Make a plot of those (x, y) for different c
- Contours are quite useful in many problems

Examples

Sum function, $g(x, y) = x + y$

Examples

Sum function, $g(x, y) = x + y$



[Click here for desmos illustration and other examples](#)

Function of two random variables

$$X, Y \sim f_{XY}, X \in \mathcal{X}, Y \in \mathcal{Y}$$

- Let $Z = g(X, Y)$ be a function of X and Y
- What is the PMF of Z ?

Function of two random variables

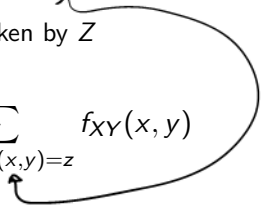
$$X, Y \sim f_{XY}, X \in \mathcal{X}, Y \in \mathcal{Y}$$

- Let $Z = g(X, Y)$ be a function of X and Y
- What is the PMF of Z ?
- Step 1: Find range of Z
 - ▶ All possible values for $g(x, y), x \in \mathcal{X}, y \in \mathcal{Y}$

Function of two random variables

$$X, Y \sim f_{XY}, X \in \mathcal{X}, Y \in \mathcal{Y}$$

- Let $Z = g(X, Y)$ be a function of X and Y
- What is the PMF of Z ?
- Step 1: Find range of Z
 - ▶ All possible values for $g(x, y), x \in \mathcal{X}, y \in \mathcal{Y}$
- Step 2: Add over the contours $g(x, y) = z$
 - ▶ Suppose z is a possible value taken by Z

$$P(Z = z) = \sum_{(x, y): g(x, y) = z} f_{XY}(x, y)$$


Example: Pair of dice, Sum

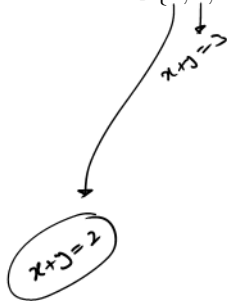
$$X, Y \sim \text{iid Unif}\{1, 2, 3, 4, 5, 6\}, Z = X + Y$$

Example: Pair of dice, Sum

$$X, Y \sim \text{iid Unif}\{1, 2, 3, 4, 5, 6\}, Z = X + Y$$

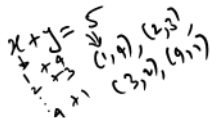
- Step 1: Find range of Z

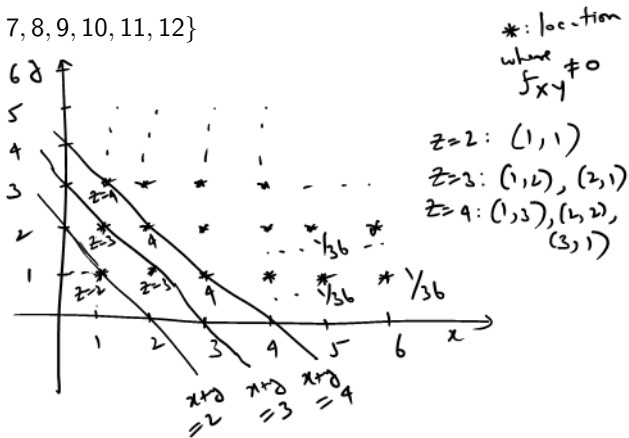
► $Z \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$



$x+y=5$

$(1,4), (2,3), (3,2), (4,1)$





Example: Pair of dice, Sum

$$X, Y \sim \text{iid Unif}\{1, 2, 3, 4, 5, 6\}, Z = X + Y$$

- Step 1: Find range of Z
 - $Z \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$
- Step 2: Add over the contours

z	2	3	4	5	6	7
	$(1,1)$	$(1,2), (2,1)$	$(1,3), (2,2), (3,1)$			$(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)$
$P(Z = z)$	1/36	2/36	3/36	4/36	5/36	6/36

z	8	9	10	11	12
	$(2,6), (3,5), (4,4), (5,3), (6,2)$	$(3,6), (4,5), (5,4), (6,3)$			$(4,6), (5,5), (6,4), (6,6)$
$P(Z = z)$	5/36	4/36	3/36	2/36	1/36

Example: Pair of dice, Max

$$X, Y \sim \text{iid Unif}\{1, 2, 3, 4, 5, 6\}, Z = \max(X, Y)$$

Example: Pair of dice, Max

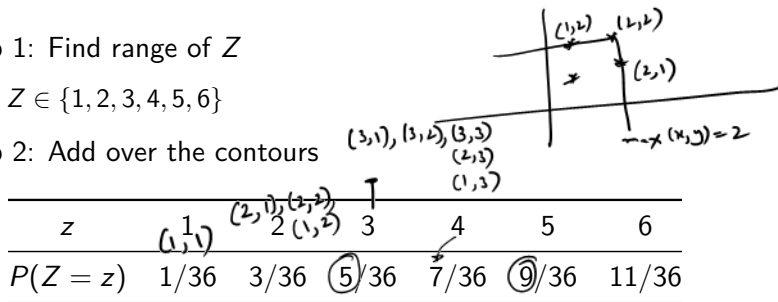
$$X, Y \sim \text{iid Unif}\{1, 2, 3, 4, 5, 6\}, Z = \max(X, Y)$$

- Step 1: Find range of Z
 - ▶ $Z \in \{1, 2, 3, 4, 5, 6\}$

Example: Pair of dice, Max

$$X, Y \sim \text{iid Unif}\{1, 2, 3, 4, 5, 6\}, Z = \max(X, Y)$$

- Step 1: Find range of Z
 - $Z \in \{1, 2, 3, 4, 5, 6\}$
- Step 2: Add over the contours



iid Uniform $\{1, \dots, n\}$: Sum

$$X, Y \sim \text{iid Unif}\{1, 2, \dots, n\}, W = X + Y$$

Step 1: range
 $\{2, 3, \dots, 2n\}$

$$W = w: \quad (1, w-1), (2, w-2), \\ \downarrow \quad \quad \quad \downarrow \\ w-1 \leq n \quad \quad \quad \underline{(w-1, 1)} \\ \quad \quad \quad \downarrow \\ \quad \quad \quad w-1 \leq n$$

iid Uniform $\{1, \dots, n\}$: Sum

$$X, Y \sim \text{iid Unif}\{1, 2, \dots, n\}, W = X + Y$$

- $W \in \{2, \dots, 2n\}$

$$P(W = w) = \begin{cases} \frac{w-1}{n^2}, & 2 \leq w \leq n+1, \\ \frac{2n-w+1}{n^2}, & n+2 \leq w \leq 2n. \end{cases}$$

iid Uniform $\{1, \dots, n\}$: Sum

$$X, Y \sim \text{iid Unif}\{1, 2, \dots, n\}, W = X + Y$$

- $W \in \{2, \dots, 2n\}$

$$P(W = w) = \begin{cases} \frac{w-1}{n^2}, & 2 \leq w \leq n+1, \\ \frac{2n-w+1}{n^2}, & n+2 \leq w \leq 2n. \end{cases}$$

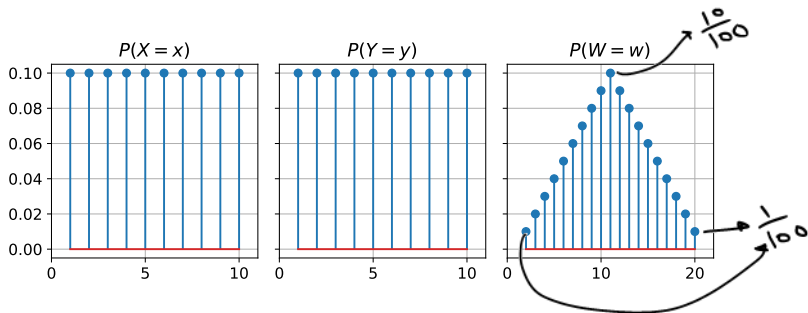


Figure 1: $P(W = w)$ vs w , $n = 10$

iid Uniform $\{1, \dots, n\}$: Max

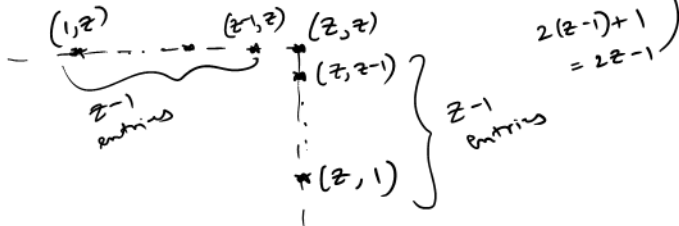
$$X, Y \sim \text{iid Unif}\{1, 2, \dots, n\}, Z = \max(X, Y)$$

iid Uniform $\{1, \dots, n\}$: Max

$$X, Y \sim \text{iid Unif}\{1, 2, \dots, n\}, Z = \max(X, Y)$$

- $Z \in \{1, \dots, n\}$

$$P(Z = z) = \frac{2z - 1}{n^2}$$



iid Uniform $\{1, \dots, n\}$: Max

$$X, Y \sim \text{iid Unif}\{1, 2, \dots, n\}, Z = \max(X, Y)$$

- $Z \in \{1, \dots, n\}$

$$P(Z = z) = \frac{2z - 1}{n^2}$$

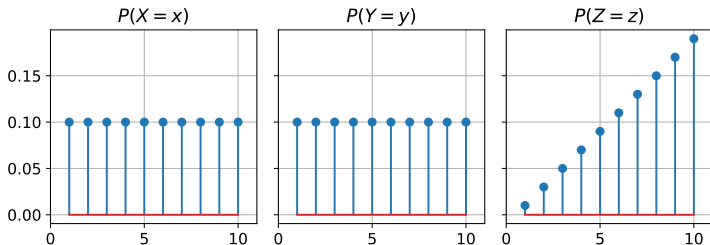


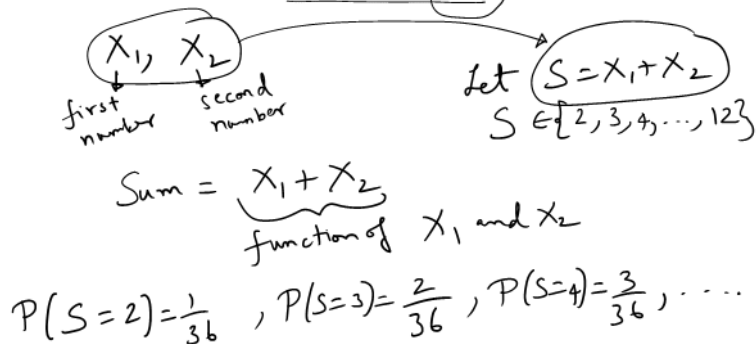
Figure 2: $P(Z = z)$ vs z , $n = 10$

Section 4

Functions of random variables

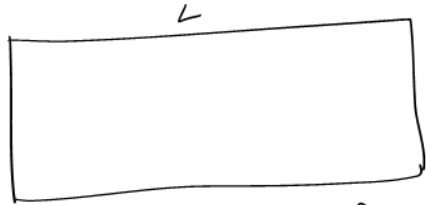
Example: Throw a die twice

A fair die is thrown twice. What is the probability that the sum of the two numbers seen is 6? What is the PMF of the sum?



Example: Area of a random rectangle

The length of a rectangle $L \sim \text{Uniform}\{5, 7, 9, 11\}$. Given $L = l$, the breadth $B \sim \text{Uniform}\{l - 1, l - 2, l - 3\}$.



$f_{L,B}$

t_1	t_2	$f_{L,B}(t_1, t_2)$	Area = <u><u>$L B$</u></u>
5	4	$\frac{1}{12}$	20
	3	$\frac{1}{12}$	15
	2	$\frac{1}{12}$	10
7	6	$\frac{1}{12}$	42
	5	$\frac{1}{12}$	35
	4	$\frac{1}{12}$	28
9	8	$\frac{1}{12}$	72
	7	$\frac{1}{12}$	63
11	6	$\frac{1}{12}$	54
	10	$\frac{1}{12}$	110
	9	$\frac{1}{12}$	99
	8	$\frac{1}{12}$	88

PMF of $g(X_1, \dots, X_n)$

Definition (PMF of a $g(X_1, \dots, X_n)$)

Suppose $X_1, \dots, X_n$ have joint PMF $f_{X_1 \dots X_n}$ with T_{X_i} denoting the range of X_i . Let $g : T_{X_1} \times \dots \times T_{X_n} \rightarrow \mathbb{R}$ be a function with range T_g . The PMF of $X = g(X_1, \dots, X_n)$ is given by

$$f_X(t) = P(g(X_1, \dots, X_n) = t) = \sum_{\underbrace{(t_1, \dots, t_n) : g(t_1, \dots, t_n) = t}_{g^{-1}(t)}} f_{X_1 \dots X_n}(t_1, \dots, t_n).$$

- Proof: write the event and use definition of joint PMF
- Directly useful for small problems
- Can be extended for joint PMF of two functions g and h

Example: Given a small table for joint PMF

t_1, t_2	$g(t_1, t_2, t_3)$	t_1, t_2, t_3	t_1	t_2	t_3	$f_{X_1 X_2 X_3}(t_1, t_2, t_3)$
0	-	0	0	0	0	1/9
0	.	1	0	0	1	1/9
0	.	2	0	0	2	1/9
1	-	2	0	1	1	1/9
2	.	3	0	1	2	1/9
0	.	1	1	0	0	1/9
0	.	3	1	0	2	1/9
0	.	2	1	1	0	1/9
1	.	3	1	1	1	1/9

① $g(X_1, X_2, X_3) = X_1 + X_2 + X_3$
 $\in \{0, 1, 2, 3\}$

→ Any other function is similar

② $X = g(X_1, X_2, X_3) = X_1 + X_2 + X_3$
 $Y = h(X_1, X_2, X_3) = X_2 X_3$

		<u>$f_{X,Y}$</u>			
		t_1, t_2, t_3			
		0	1	2	3
t_1, t_2	0	1/9	2/9	2/9	1/9
	1	0	0	1/9	1/9
	2	0	0	0	1/9

Working

Example: Binomial from Bernoulli trials

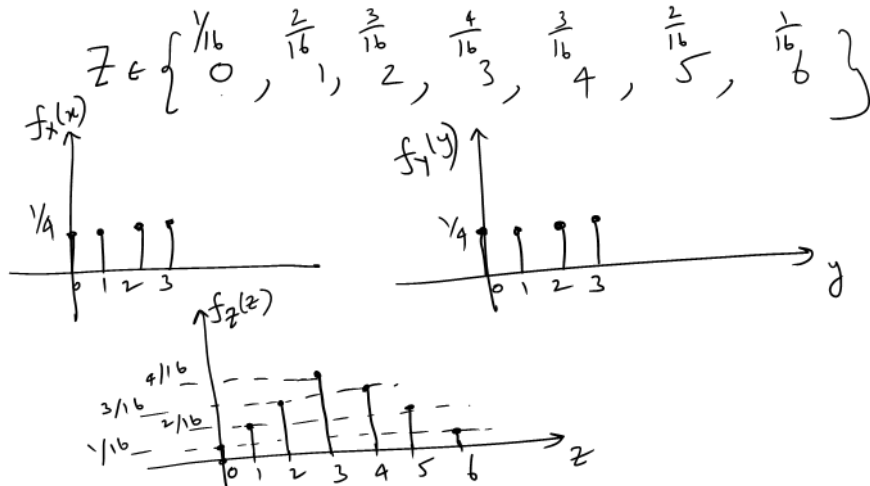
Let $X_1, \dots, X_n$ be the results of n i.i.d. $\text{Bernoulli}(p)$ trials. The sum of the n random variables $X_1 + \dots + X_n$ is $\text{Binomial}(n, p)$.


of successes
in n indep Bernoulli trials

→ Sum of n indep $\text{Bernoulli}(p) = \text{Binomial}(n, p)$

Example: Sum of two uniforms

Let $X \sim \text{Uniform}\{0, 1, 2, 3\}$ and $Y \sim \text{Uniform}\{0, 1, 2, 3\}$ be independent. Find the PMF of $Z = X + Y$.



Sums of two random variables taking integer values

Suppose X and Y take integer values and let their joint PMF be f_{XY} . Let $Z = X + Y$.

Sums of two random variables taking integer values

Suppose X and Y take integer values and let their joint PMF be f_{XY} . Let $Z = X + Y$.

Let z be some integer.

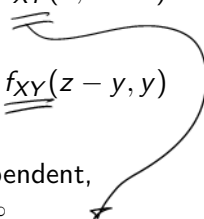
$$\begin{aligned} P(Z = z) &= P(X + Y = z) \\ &= \sum_{x=-\infty}^{\infty} P(X = x, Y = z - x) \\ &= \sum_{x=-\infty}^{\infty} f_{XY}(x, z - x) \\ &= \sum_{y=-\infty}^{\infty} f_{XY}(z - y, y) \end{aligned}$$

Handwritten annotations:
- An arrow points from the z in $P(X + Y = z)$ to the $z - x$ in $P(X = x, Y = z - x)$.
- The summation index x is circled, with an arrow pointing to the text "any integer".

Sums of two random variables taking integer values

Suppose X and Y take integer values and let their joint PMF be f_{XY} . Let $Z = X + Y$.

Let z be some integer.

$$\begin{aligned}P(Z = z) &= P(X + Y = z) \\&= \sum_{x=-\infty}^{\infty} P(X = x, Y = z - x) \\&= \sum_{x=-\infty}^{\infty} f_{XY}(x, z - x) \\&= \sum_{y=-\infty}^{\infty} f_{XY}(z - y, y)\end{aligned}$$


Convolution: If X and Y are independent,

$$f_{X+Y}(z) = \sum_{x=-\infty, \text{integer}}^{\infty} f_X(x) f_Y(z - x)$$

Sum of two independent Poissons

Let $X \sim \text{Poisson}(\lambda_1)$ and $Y \sim \text{Poisson}(\lambda_2)$ be independent.

① Find the PMF of $Z = X + Y$.

② Find the conditional distribution of $X|Z$.

① $z \in \{0, 1, 2, \dots\}$

$$f_Z(z) = \sum_{x=0}^z f_X(x) \cdot f_Y(z-x)$$

if $x < 0$ if $x > z$

$$= \sum_{x=0}^z e^{-\lambda_1} \frac{\lambda_1^x}{x!} \cdot e^{-\lambda_2} \frac{\lambda_2^{z-x}}{(z-x)!}$$

$$= \frac{e^{-\lambda_1} e^{-\lambda_2}}{z!} \left(\sum_{x=0}^z \frac{z!}{x!(z-x)!} \lambda_1^x \lambda_2^{z-x} \right)$$

$\parallel$

$$f_Z(z) = \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^z}{z!}$$

$Z \sim \text{Poisson}(\lambda_1 + \lambda_2)$

② $P(X=k|Z=n) = \frac{P(X=k, Z=n)}{P(Z=n)} = \frac{P(X=k) \cdot P(Z=n|X=k)}{P(Z=n)} = \frac{P(X=k) P(Y=n-k)}{P(Z=n)}$

$X|Z \sim \text{Binomial}(n, \frac{\lambda_1}{\lambda_1 + \lambda_2})$

$Y|Z \sim \text{Binomial}(n, \frac{\lambda_2}{\lambda_1 + \lambda_2})$

$$= \frac{e^{-\lambda_1} \lambda_1^k / k! \cdot e^{-\lambda_2} \lambda_2^{n-k} / (n-k)!}{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n / n!} = \frac{n!}{k!(n-k)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-k}$$

Functions and independence

- If X and Y are independent, $g(X)$ and $h(Y)$ are independent for any two functions g and h

Functions and independence

- If X and Y are independent, $g(X)$ and $h(Y)$ are independent for any two functions g and h
- If X_1, X_2, X_3, X_4 are mutually independent,

Functions and independence

- If X and Y are independent, $g(X)$ and $h(Y)$ are independent for any two functions g and h
- If X_1, X_2, X_3, X_4 are mutually independent,
 - ▶ $g(X_1, X_2)$ is independent of $h(X_3, X_4)$

Functions and independence

- If X and Y are independent, $g(X)$ and $h(Y)$ are independent for any two functions g and h
- If X_1, X_2, X_3, X_4 are mutually independent,
 - ▶ $g(X_1, X_2)$ is independent of $h(X_3, X_4)$
 - ▶ $g(X_1, X_2, X_3)$ is independent of $h(X_4)$

Functions and independence

- If X and Y are independent, $g(X)$ and $h(Y)$ are independent for any two functions g and h
- If X_1, X_2, X_3, X_4 are mutually independent,
 - ▶ $g(X_1, X_2)$ is independent of $h(X_3, X_4)$
 - ▶ $g(X_1, X_2, X_3)$ is independent of $h(X_4)$
- Functions of non-overlapping sets of independent random variables are also independent.

Exercises

- Sum of independent $\text{Binomial}(m, p)$ and $\text{Binomial}(n, p)$
- Sum of independent $\text{Geometric}(p)$ and $\text{Geometric}(q)$
- Sum of r i.i.d. $\text{Geometric}(p)$
- Sum of independent $\text{Negative-Binomial}(r, p)$ and $\text{Negative-Binomial}(s, p)$

Minimum of two random variables

$$X, Y \sim f_{XY}$$

$$Z = \min(X, Y): \text{function of } X \text{ and } Y$$

Ex: Throw a die twice. $Z = \text{minimum of the two numbers seen.}$

Minimum of two random variables

$$X, Y \sim f_{XY}$$

$Z = \min(X, Y)$: function of X and Y

Ex: Throw a die twice. Z = minimum of the two numbers seen.

$$\begin{aligned} f_Z(z) &= P(\min(X, Y) = z) \\ &= P((X = z \text{ and } Y = z) \text{ or } (X = z \text{ and } Y > z) \\ &\quad \text{or } (X > z \text{ and } Y = z)) \\ &= f_{XY}(z, z) + \sum_{t_2 > z} f_{XY}(z, t_2) + \sum_{t_1 > z} f_{XY}(t_1, z) \end{aligned}$$

Minimum of two random variables

$$X, Y \sim f_{XY}$$

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What about maximum?

Example: Throw a die twice

Throw a die twice. Z = minimum of the two numbers seen.

	1	2	3	4	5	6
1	1/36	1/36	1/36	1/36	1/36	1/36
2	1/36	1/36	1/36	1/36	1/36	1/36
3	1/36	1/36	1/36	1/36	1/36	1/36
4	1/36	1/36	1/36	1/36	1/36	1/36
5	1/36	1/36	1/36	1/36	1/36	1/36
6	1/36	1/36	1/36	1/36	1/36	1/36

$$P(Z=1) = \frac{11}{36} \quad P(Z=2) = \frac{9}{36} \quad P(Z=3) = \frac{7}{36}$$

$$P(Z=4) = \frac{5}{36} \quad P(Z=5) = \frac{3}{36} \quad P(Z=6) = \frac{1}{36}$$

Independent case: cumulative distribution function (CDF) of maximum

Definition (CDF of a random variable)

Cumulative distribution function of a random variable X is a function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined as

$$F_X(x) = P(X \leq x).$$

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Suppose X and Y are **independent** and $Z = \max(X, Y)$.

$$\begin{aligned} F_Z(z) &= P(\max(X, Y) \leq z) \\ &= P((X \leq z) \text{ and } (Y \leq z)) \\ &= P(X \leq z)P(Y \leq z) \quad \text{by independence} \\ &= F_X(z)F_Y(z) \end{aligned}$$

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Independent: CDF of maximum is product of CDFs. What about min?

Problem: Min and max of i.i.d. sequences

Let $X_1, \dots, X_n \sim \text{i.i.d. } X$. Find the distribution of the following:

① $\min(X_1, \dots, X_n)$

② $\max(X_1, \dots, X_n)$

$$\textcircled{2} \quad \mathbb{P}(\max(X_1, \dots, X_n) \leq z) = \mathbb{P}(X_1 \leq z, X_2 \leq z, \dots, X_n \leq z) \\ = (\mathbb{P}(X \leq z))^n = (F_X(z))^n$$

$$\textcircled{1} \quad \mathbb{P}(\min(X_1, \dots, X_n) \geq z) = \mathbb{P}(X_1 \geq z, X_2 \geq z, \dots, X_n \geq z) \\ = (\mathbb{P}(X \geq z))^n$$

Problem: Min of two independent Geometrics

Let $X \sim \text{Geometric}(p)$ and $Y \sim \text{Geometric}(p)$ be independent. Find the distribution of $\min(X, Y)$.

$$P(X=k) = (1-p)^{k-1} \cdot p$$

$$\begin{aligned} P(\underline{\min(X, Y)} \geq k) &= P(X \geq k) \cdot P(Y \geq k) \\ &= (1-p)^{k-1} (1-p)^{k-1} = ((1-p)^2)^{k-1} \end{aligned}$$

$$P(\min(X, Y) \geq k+1) = ((1-p)^2)^k \quad q = (1-p)^2$$

$$\begin{aligned} P(\underline{\min(X, Y)} = k) &= P(\min(X, Y) \geq k) - P(\min(X, Y) \geq k+1) \\ &= q^{k-1} - q^k = q^{k-1} (1-q) \end{aligned}$$

$$\min(X, Y) \sim \text{Geometric}(1-q)$$

$$X_1 \sim \text{Geometric}(p_1), X_2 \sim \text{Geometric}(p_2) \text{ indep}$$

$$\min(X_1, X_2) \sim \text{Geometric}(1 - (1-p_1)(1-p_2))$$

$\rightarrow p_1 + p_2 - p_1 p_2$